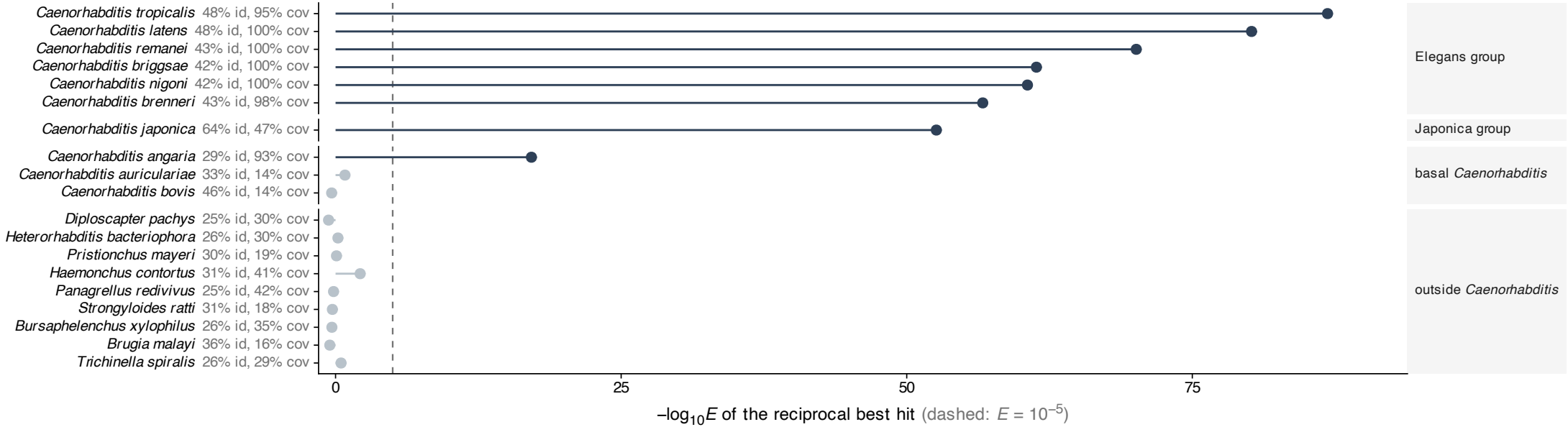


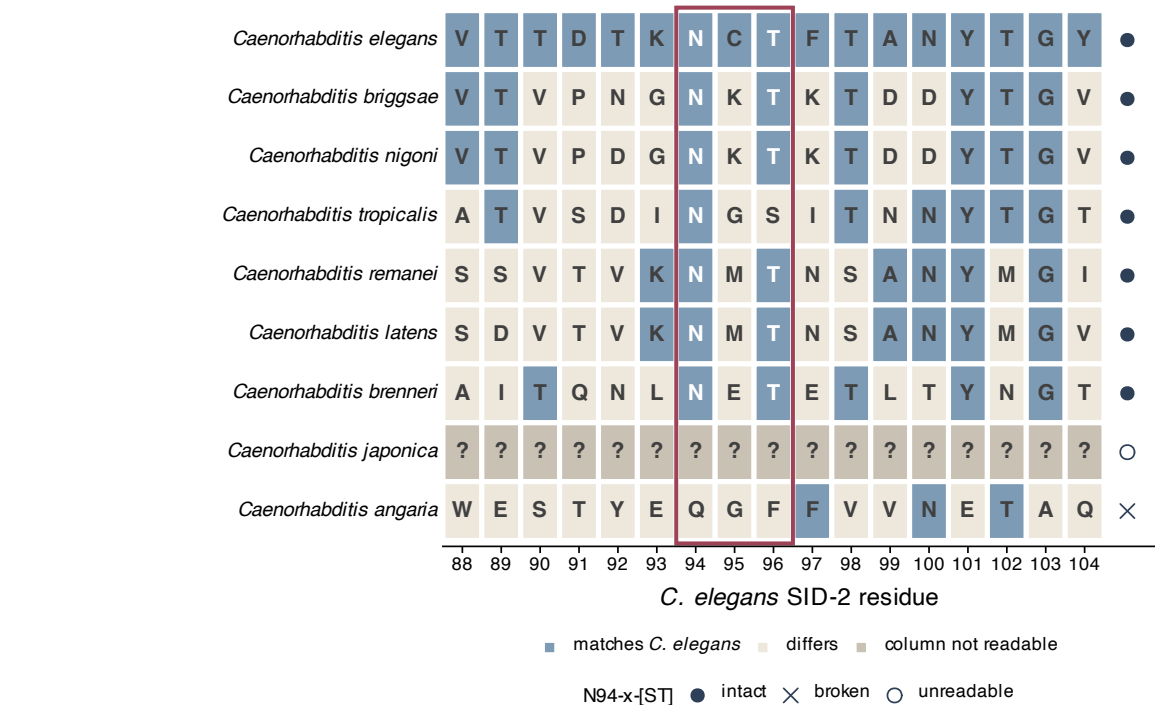
A Outside *Caenorhabditis*, SID-2 has no detectable ortholog

Reciprocal-best-hit blastp of *C. elegans* SID-2 against twenty UniProt reference proteomes, 488,718 proteins. The query itself is not drawn, so nineteen comparators are: eight clear $E < 1e-5$ and every one of them is a *Caenorhabditis*. Nothing beyond the genus clears it, including *Diploscapter pachys*, the sister genus, at $E = 4.1$. Two *Caenorhabditis* proteomes also fail, which is weak evidence about the gene and may only say their annotation is incomplete.



B The sequon holds across the Elegans group and breaks at the base of the genus

Residues aligned to *C. elegans* 88-104. The shaded columns are the N94-x-T96 sequon. Its two constrained positions hold while the middle one varies freely; *C. angaria* has Q-G-F. *japonica*'s ortholog is split across two proteome entries and its N-terminal fragment aligns into a TTDT repeat offset by two to three residues, so that row is drawn as unreadable rather than as substitutions.



C Constrained positions high, free one lowest

All 173 ectodomain positions (21-193), binned by how many of the 6 orthologs share the *C. elegans* residue. The background is 2.23 of 6 (37%) and only 15% of positions are conserved in all. Calibration that cuts the other way: three of the five ectodomain sequons are intact in all 6 orthologs, so a fully conserved sequon is the norm in this protein and the one at 94 is not a standout.

